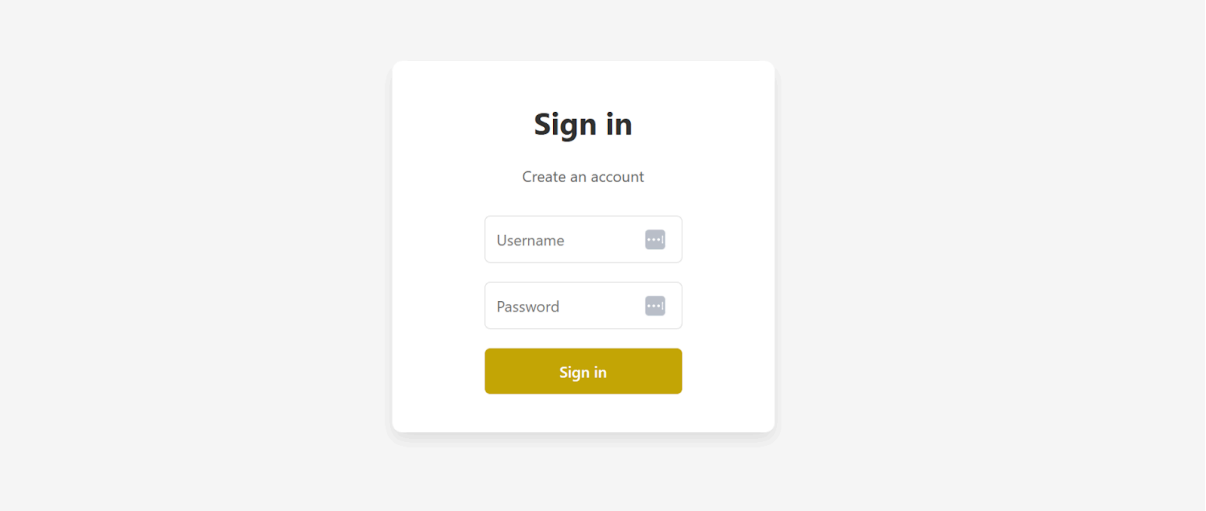
Part 1.1: End User Guide

Section 1: Getting Started

1.1 Accessing the Web Application

1. Open your web browser (Chrome, Firefox, Safari, or Edge recommended).
2. Navigate to the Australian Weather Explorer application URL.
3. You will be directed to the login page if you're not already logged in.

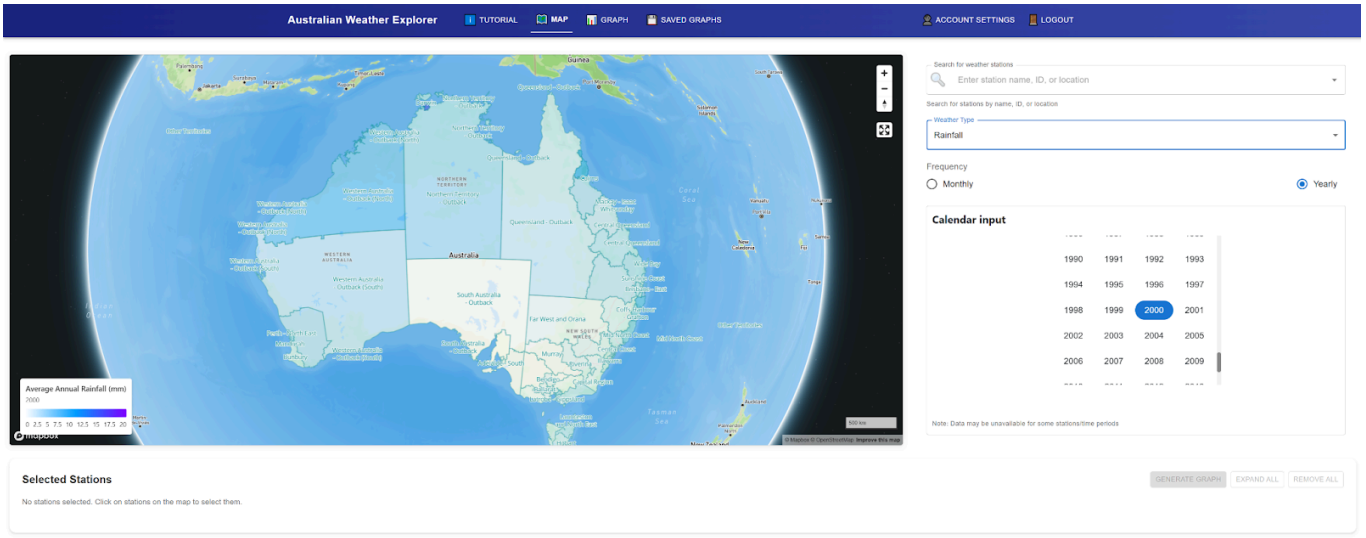
1.2 Creating an Account and Logging In



1. If you don't have an account, click the "Sign Up" button on the login page.
2. Fill in the required information:
    - Username
    - Email address
    - Password
3. Click "Sign Up" to create your account.
4. Once registered, log in with your credentials on the login page.
5. Enter your username/email and password.
6. Click "Login" to access the application.

Section 2: Main Interface Overview

2.1 Dashboard Layout



The main dashboard consists of several key areas:

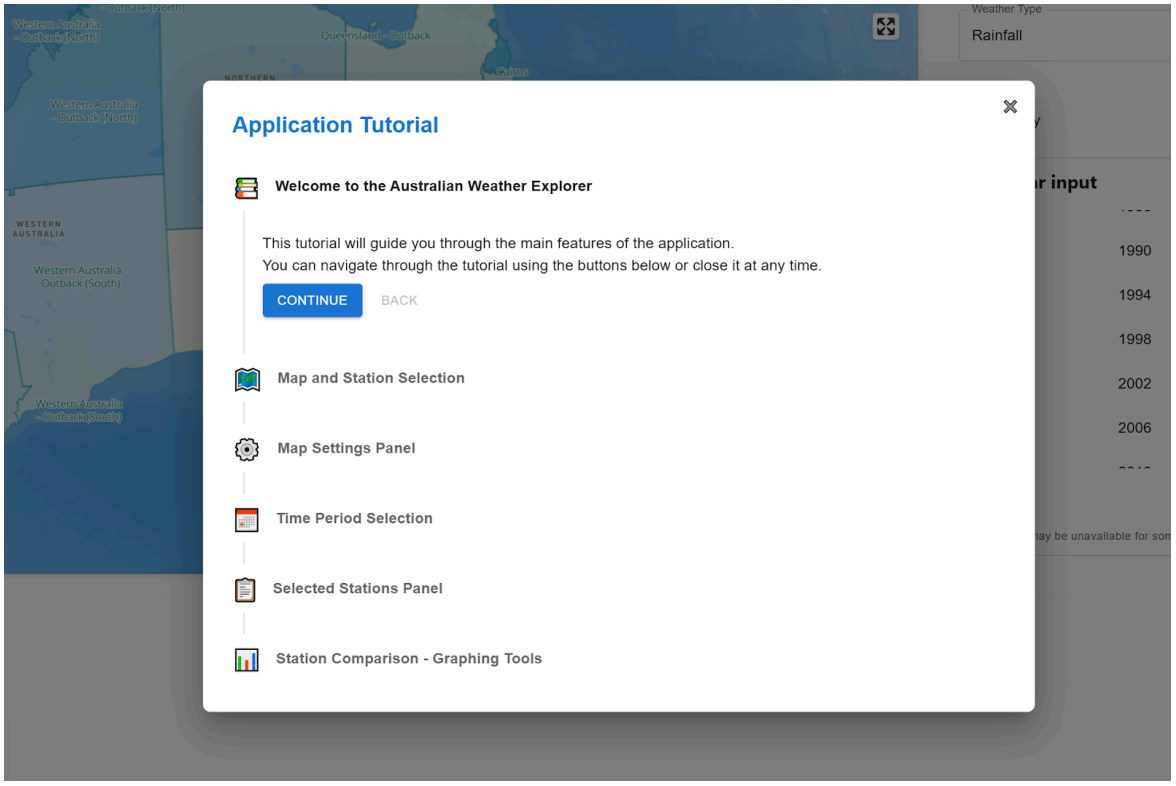
- Navigation bar (top) - Access different sections of the application.
- Map area (left side) - Interactive Choropleth map of Australia showing weather stations.
- Sidebar (right side) - Control panel for map settings and filters.
- Selected Stations Panel (bottom) - Shows your currently selected weather stations and detailed weather information for the time period selected (month or year).

2.2 Navigation Bar

The navigation bar provides access to the main sections of the application:

- Tutorial - A guide for new users.
- Home/Map - The main dashboard with the interactive map and station information.
- Graph - Generate and compare data between selected stations.
- Saved Graphs - View and manage your saved data visualisations.
- Account - Access your account settings and preferences.
- Logout - Securely log out of the application.

2.3 Tutorial



The Australian Weather Explorer includes an interactive tutorial designed to help new users quickly understand and navigate the application's features. This guided walkthrough is particularly valuable for first-time users who want to learn the system efficiently. The tutorial can be accessed at any time from the Navigation Bar.

The tutorial is organised into several key sections that correspond to the main functionalities of the application:

- 🗺️ Map and Station Selection
- ⚙️ Map Settings Panel
- 📅 Time Period Selection
- 📋 Selected Stations Panel
- 📊 Station Comparison - Graphing Tools

When logging in for the first time, the tutorial will automatically appear, introducing the user to the Australian Weather Explorer. It is recommended that new users go through the tutorial before exploring the application.

Section 3: Map and Weather Stations

3.1 Map

3.1.1 Exploring the Map

The interactive map provides a comprehensive view of weather data across Australia. To navigate effectively:

- Zooming: Use your mouse scroll wheel to zoom the map in and out.
- Panning: Click and hold while dragging the map to reposition your view and explore different areas of Australia.
- Clicking: This action is used to select regions and stations. This is explained in section 3.2.1

3.1.2 Map Control Panel

The map control panel offers powerful visualisation options to analyse different weather metrics:

- **Data Type Selection:** Choose from three key weather variables to display on the Choropleth Map:
  - Rainfall
  - Minimum Temperature
  - Maximum Temperature
- **Regional Display:** The map divides Australia into Statistical Area 4 (SA4) regions as defined by the Australian Bureau of Statistics. Each region displays averaged data from all weather stations located within its boundaries.
- **Time Period Controls:** Customise your data view by selecting:
  - Frequency: Toggle between monthly or yearly data aggregation.
  - Time Period: Select a specific month and year, or view annual data for a particular year.

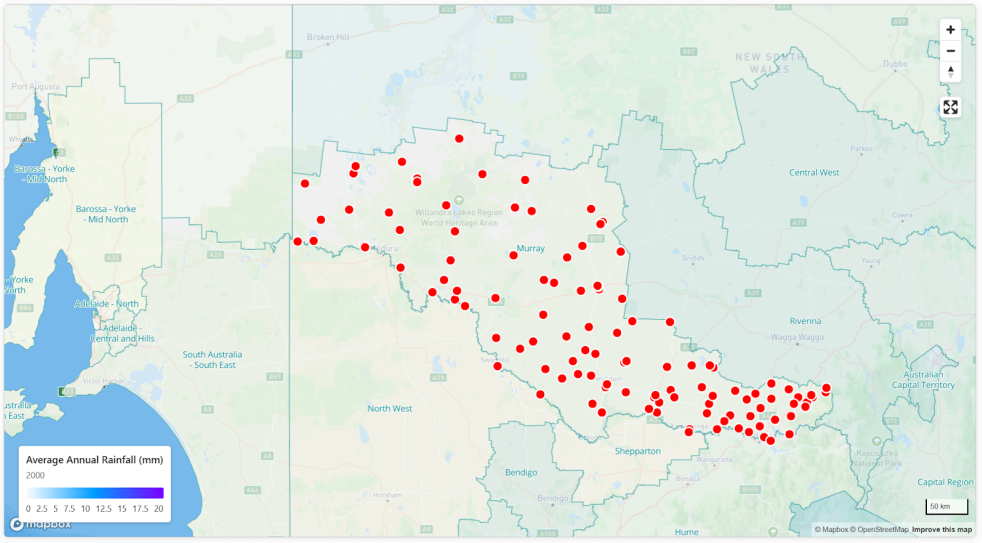
This visualisation approach allows you to quickly identify regional weather patterns and compare conditions across different parts of Australia at a glance.

3.2 Working with Weather Stations

3.2.1 Selecting Weather Stations

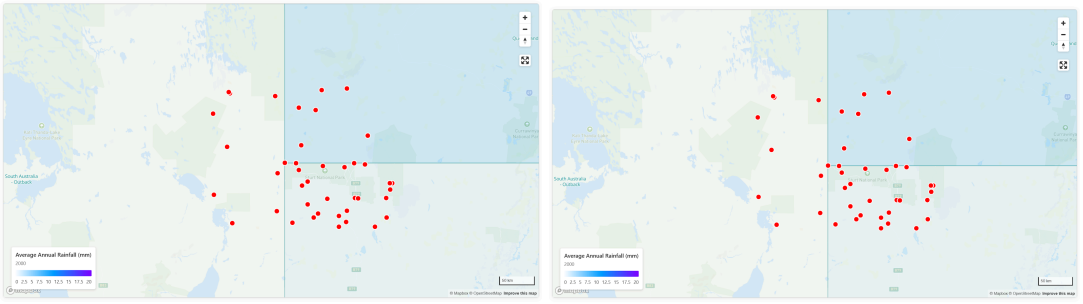
To select a station from the Map:

1. Click on a SA4 region on the map. This will add a red marker to each station in the area.

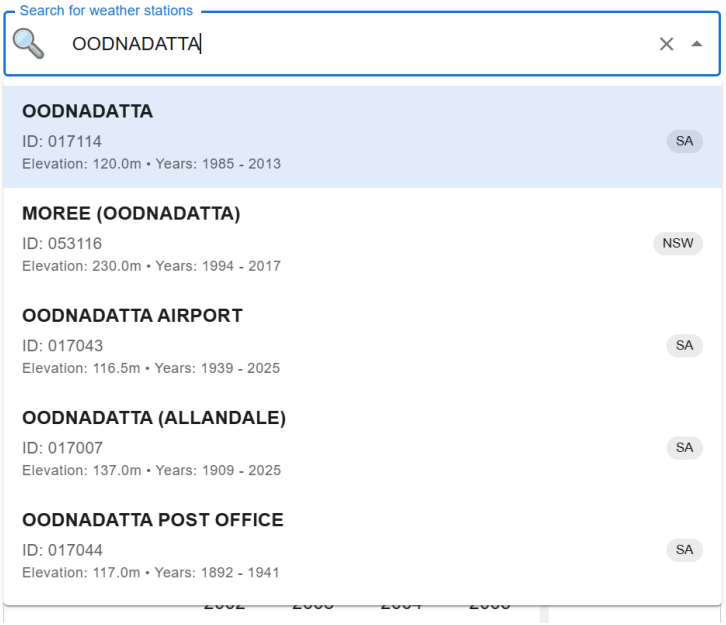


To unhighlight a region, click on the same region again. You can also click on another region to select it instead.

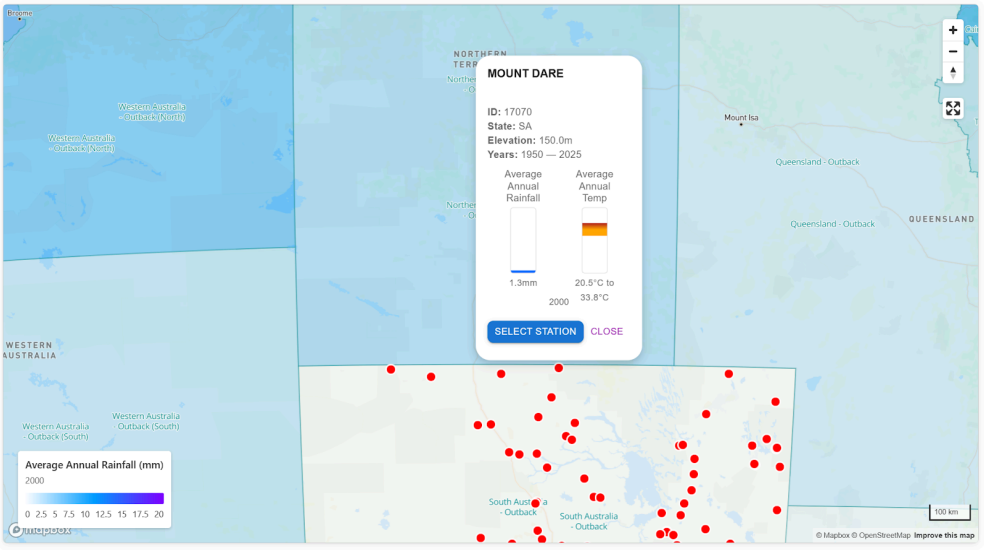
You can also highlight the stations within a specific area by holding down the Shift key and dragging a rectangle across the map.



Alternatively you can search for a specific station using the search bar in the Map Control Panel. This will pop up the station card on the map.



2. Click on a station to view its basic information such as its name and weather data for the date selected.



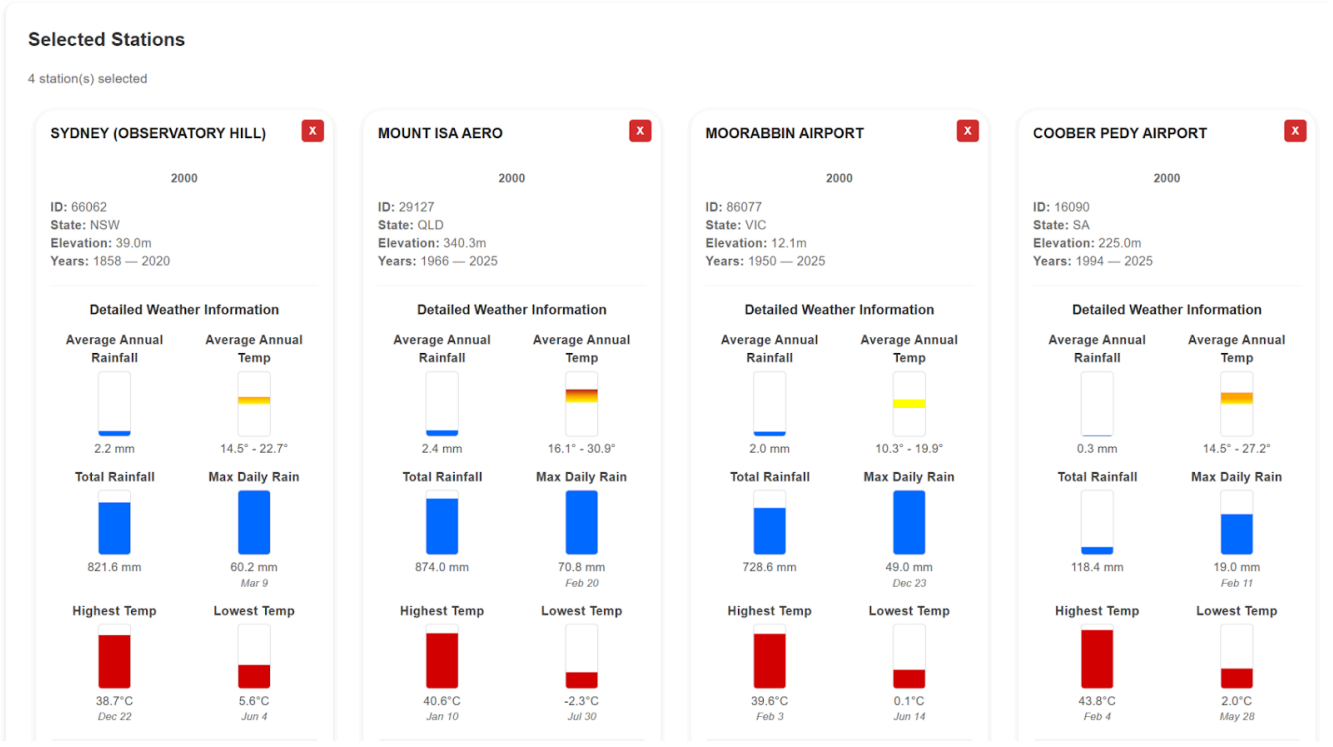
3. Click on the Select Station button to add it to the selected stations.

3.2.2 Managing Selected Stations

- View your selected stations in the panel at the bottom of the screen.
- To remove a station, click the "X" button next to its name.
- To clear all selected stations, click the "Remove All" button.

Section 4: Analysing Weather Data

4.1 Comparing Weather Station Information



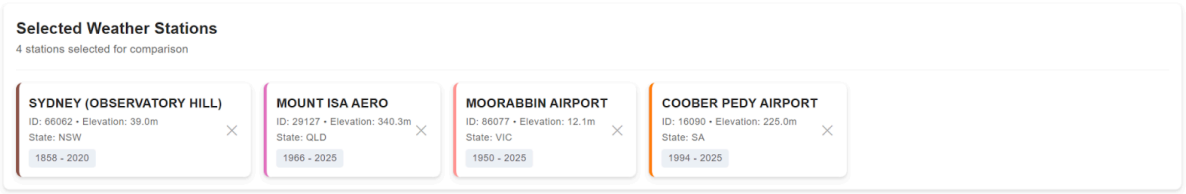
1. Navigate to the "Selected Stations" section under the map. This is the comparison section.
2. The comparison section displays data for all selected stations in cards.
    - The data displayed corresponds to the date period written at the top of the cards. This time period is linked to the same time period selected for the map in the Map Control Panel.
    - Like the Choropleth Map, this data is aggregated either by month or year.

Click on the “Advanced Statistics” button to see more weather information about the station.

4.2 Graphing Weather Station Information

Once two or more stations have been selected, click on the "Generate Graph" button in the comparison section. Alternatively, you can navigate to the graphing page by clicking "Graph" in the Navigation Bar. This page consists of four main sections:

4.2.1 Selected Weather Stations



This section displays all weather stations currently selected for comparison.

For each station, you can view:

- Station name
- Station ID number
- Elevation (in meters)
- State location
- Years of available data
- Assigned colour for visualisation (automatically generated for each station)

To remove a station from your comparison, click the "×" button next to the station you wish to remove.

4.2.2 Graph Settings

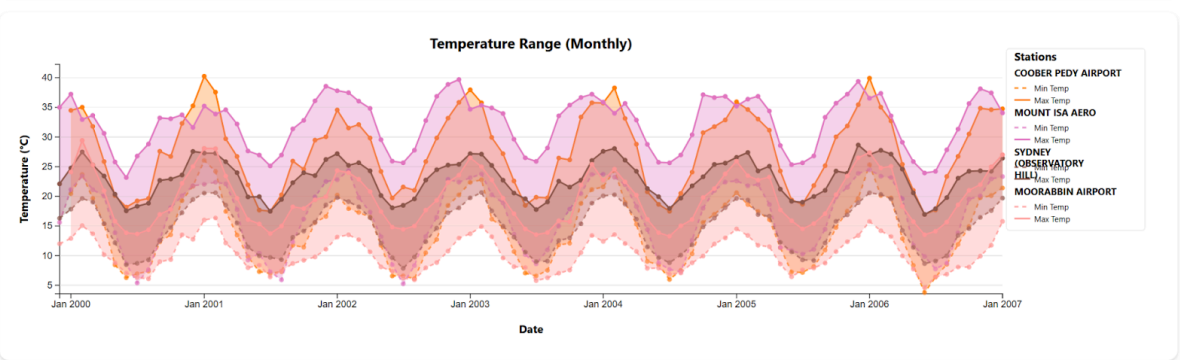


This section provides comprehensive controls to customise your weather data visualisation:

1. **Weather Type: Select the specific weather metric you wish to analyse:**
    - Rainfall
    - Maximum Temperature
    - Minimum Temperature
    - Temperature Range (shows both minimum and maximum temperatures)
2. **Frequency (Choose how data should be aggregated):**
    - Daily (default)
    - Monthly
    - Yearly
3. **Date Range: (Set your desired time period for analysis):**
    - Start: Use the calendar selector to choose when your analysis should begin.
    - End: Use the calendar selector to choose when your analysis should end.

⚠️ The system will not limit options to periods where data is available for all selected stations. This allows you to graph extended periods of time where data may be available for a limited number of selected stations. When data for a station is available it will be automatically added to the graph for the available time period.

4.2.3 Graph Display



The main graph area dynamically updates based on your selected settings, showing:

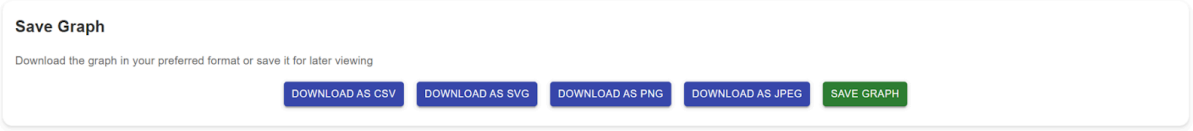
- X-axis: Timeline according to your selected frequency and date range.
- Y-axis: Scale appropriate to your selected weather type (e.g., degrees Celsius for temperature).
- Legend: Colour-coded identification of each station with separate indicators for minimum and maximum values when applicable.
- Data visualisation: Weather patterns displayed using line charts.

Example use cases of the graphing tool:

- Identify seasonal patterns across different locations.
- Compare temperature variations between coastal and inland stations.
- Analyse climate differences based on elevation or geographical location.
- Detect seasonal trends or long-term trends over multiple years.

To export or save your graph for later reference, use the "Save Graph" option in the top-right corner of the visualisation panel.

4.2.4 Downloading and Saving Graphs



The Australian Weather Explorer offers multiple options for preserving your weather data visualisations for future reference or external use.

Download Options:

You can export your generated graphs in several formats by clicking the download icon in the upper right corner of the graph display:

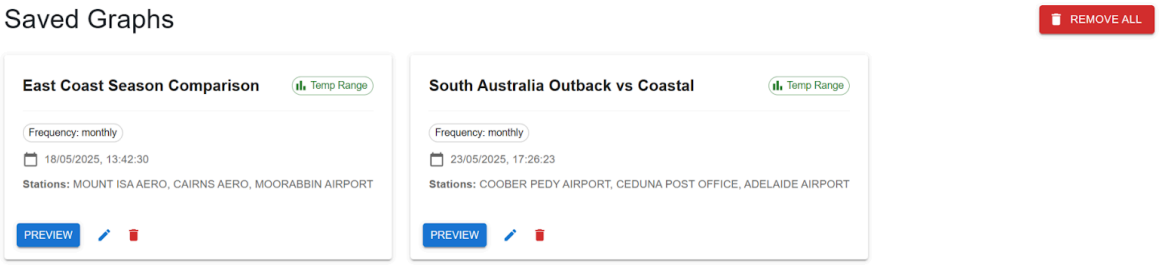
- PNG/JPEG Image: Download a high-resolution image file suitable for presentations or reports.
- SVG: Download a scalable vector graphic for professional editing or publication.
- CSV Data: Export the raw numerical data for all selected stations for further analysis in spreadsheet software.

Each download option provides appropriate resolution and formatting for different use cases, from quick sharing to detailed analysis.

Saving to Your Account:

To save a graph configuration to your user account for future access click the green "Save Graph" button located below the graph display.

Accessing Saved Graphs

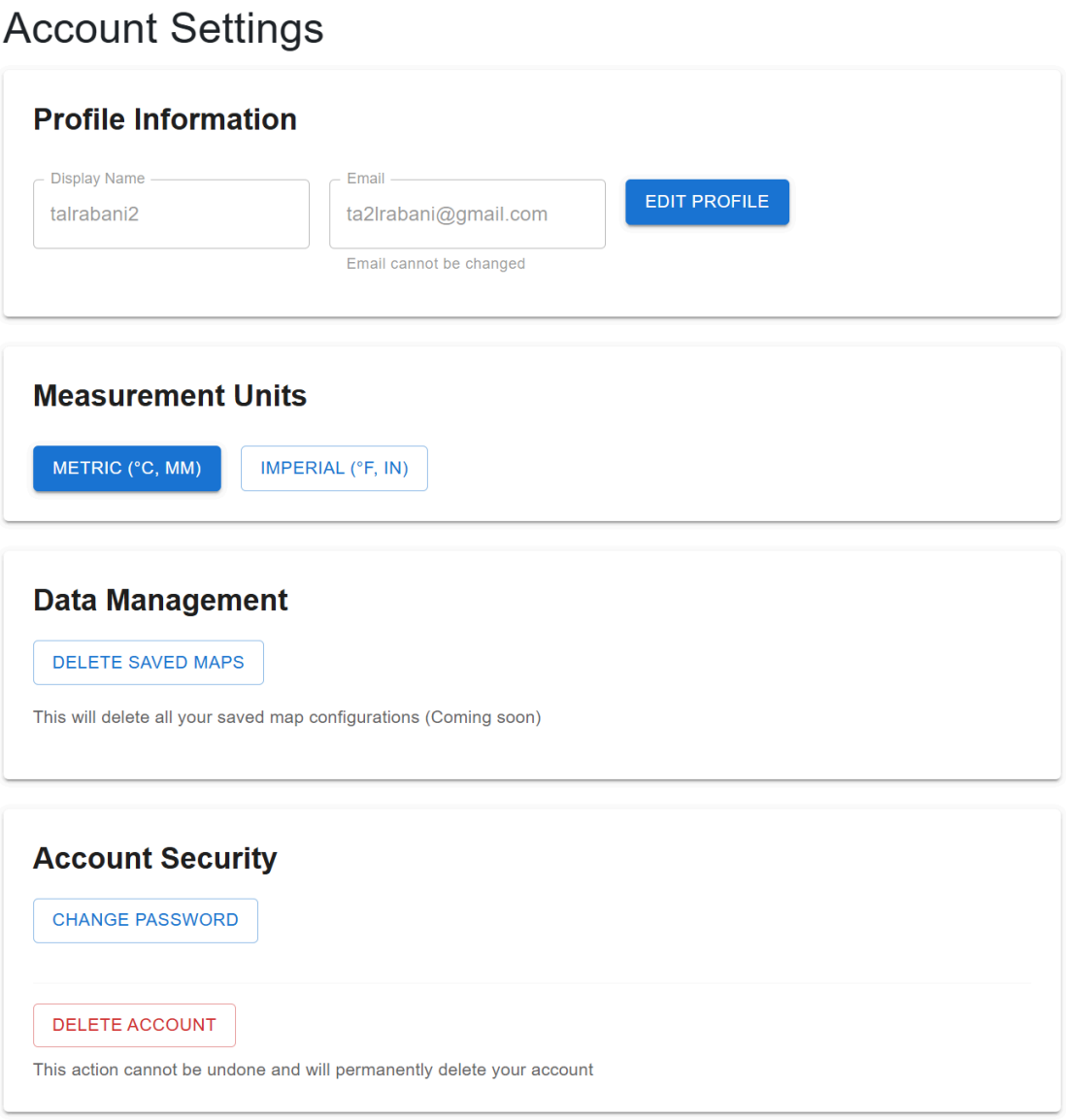


You can access all your previously saved graphs by:

1. Navigating to the "Saved Graphs" section from the main navigation menu.
2. Browsing your saved graph library, sorted by date created.
3. Click on any saved graph to:
   - View.
   - Rename.
   - Download.
   - Delete it if no longer needed

The saved graphs feature allows you to build a personal library of weather visualisations that you can revisit.

Section 5: Account Management



5.1 Updating Profile Information

1. Navigate to the "Account" page from the navigation menu.
2. Click Edit Profile.
3. Update your Display Name.
4. Click "Save Changes" to update your profile

5.2 Changing Password

1. On the "Account" page, go to the "Security" section
2. Click Change Password.
3. Enter your current password
4. Enter and confirm your new password.
5. Click "Change Password" to confirm the update.

5.3 Logging Out

1. Click "Logout" from the Navigation Bar.
2. You will be redirected to the login page.

Section 6: Common Issues and Solutions

- **Choropleth Map is only partially coloured:** Adjust the frequency and time period to a more recent date. Data before 1920 is very limited.
- **Station only displays rainfall data but no temperature data:** Most BOM stations do not include temperature data. Both rainfall and temperature data is only available for some stations. Tip: Generally airport weather stations will contain both data.
- **Graph is empty:** This will occur if all of the stations selected have no data for the type and time period selected. Adjust the time period or data type being displayed. If many stations are selected then data may take longer to load. Please allow up to 30 seconds for large data sets.

Part 1.2: Technical Guide

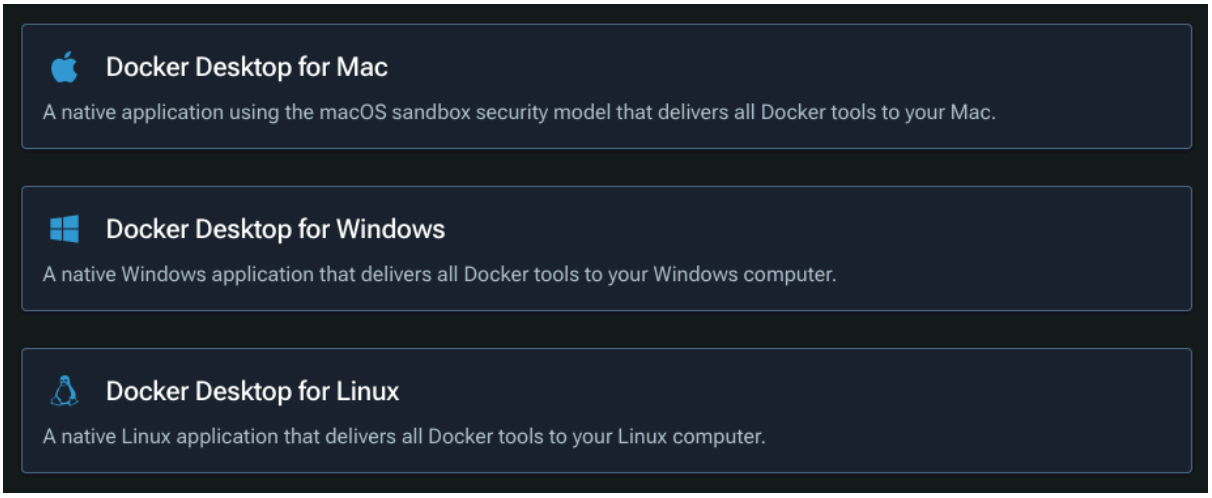
Section 1: Installing necessary software

1.1 Installing Docker

Docker is required to containerise the client, server and database of the application. It can be installed using the following link:

https://docs.docker.com/get-started/get-docker/

Select and follow the installation for the operating system relevant to you:



1.2 Installing Git

Git is required to install the application repository. Proceed to the following link:

https://git-scm.com/downloads

Select and follow the installation for the operating system relevant to you:



1.3 Installing Python

Python is required to extract the BOM rainfall and temperature data. To install Python, proceed to the following link:

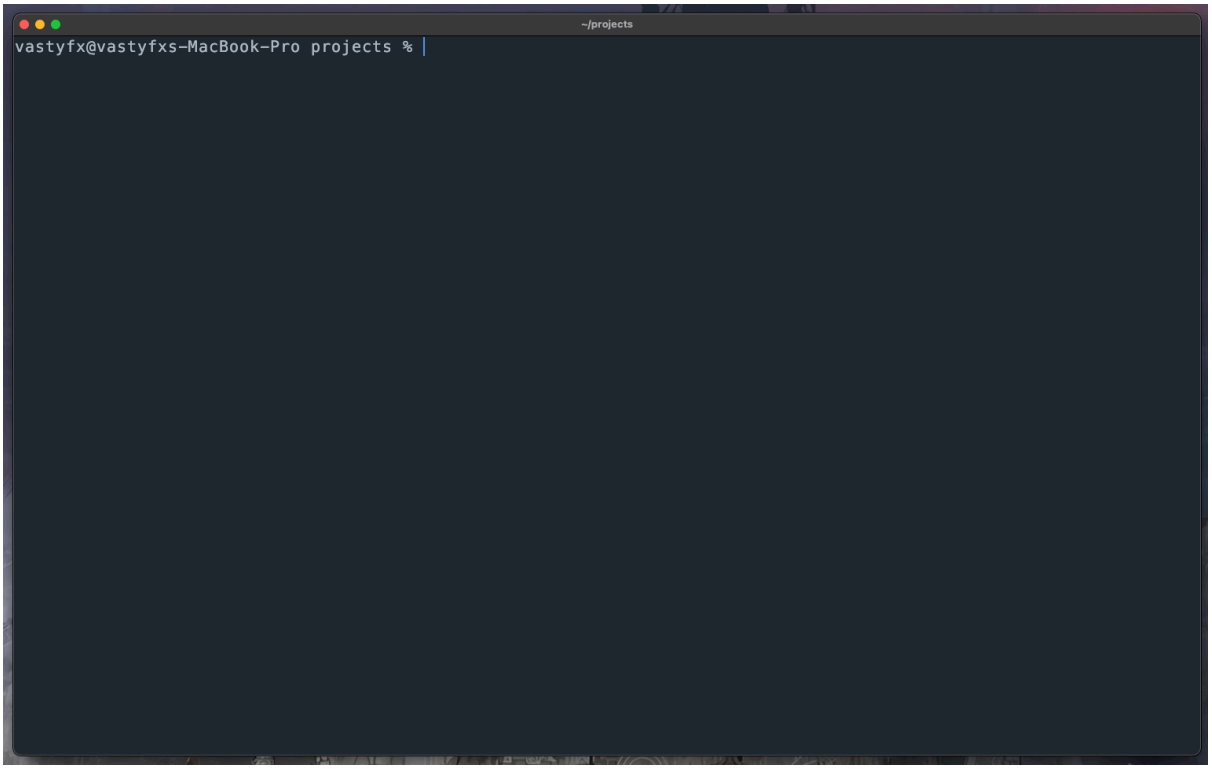
https://www.python.org/downloads/

Install in accordance with your relevant operating system:



1.4 Installing Repository

Open your terminal on your computer. Ensure you are located in the directory you would like to install the source code of the application:



Execute the command:

`git clone https://github.com/talrabani/FIT3161-3162-Senior-Project.git`

If installation is successful, you will see the following response:

```
vastyfx@vastyfxs-MacBook-Pro test % git clone https://github.com/talrabani/FIT3161-3162-Senior-Project.git
Cloning into 'FIT3161-3162-Senior-Project'...
remote: Enumerating objects: 24431, done.
remote: Counting objects: 100% (99/99), done.
remote: Compressing objects: 100% (48/48), done.
remote: Total 24431 (delta 66), reused 52 (delta 51), pack-reused 24332 (from 2)
Receiving objects: 100% (24431/24431), 1004.65 MiB | 12.53 MiB/s, done.
Resolving deltas: 100% (1748/1748), done.
Updating files: 100% (21693/21693), done.
```

Section 2: Extracting rainfall and weather data

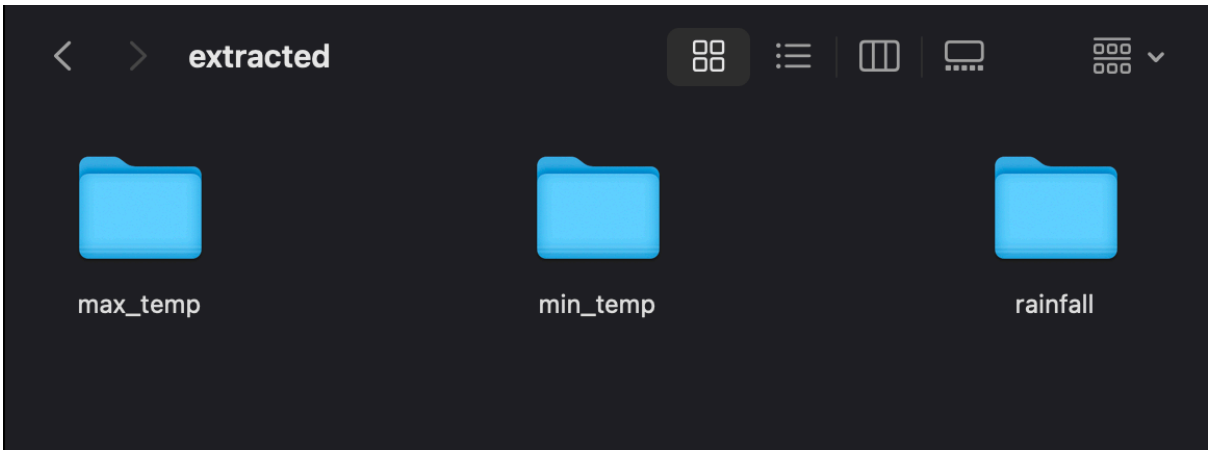
2.1 Extracting the data

All BOM rainfall and weather data was installed during installation of the repository, however this data is currently compressed and needs to be extracted for the application to process.

In your terminal, navigate to the root directory of the application. Execute the following command:

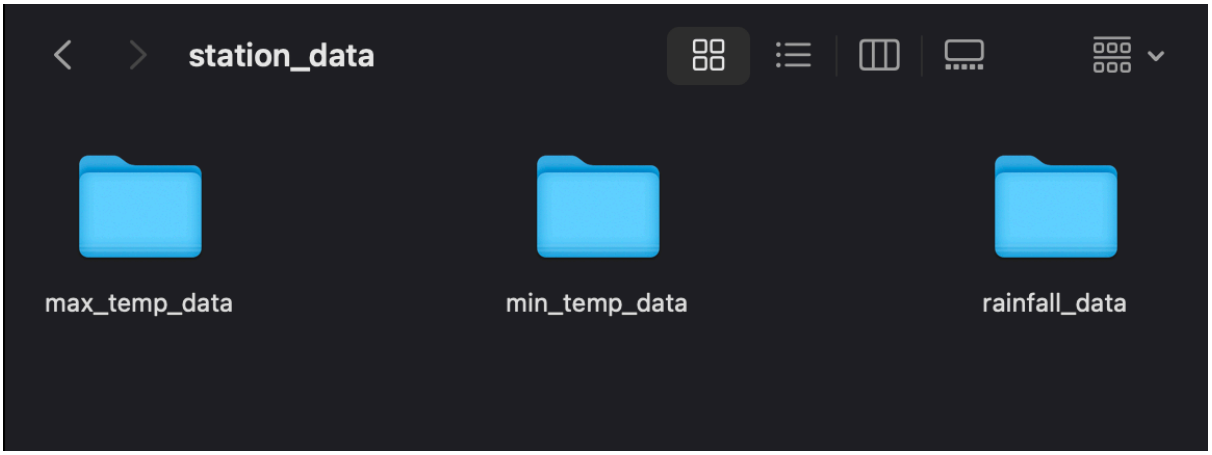
```
python3 db/extract_zips.py
```

This will extract all data to a db/extracted/. You should see the following folders inside, populated with the relevant csv files:



2.2 Relocating the extracted files

In db/, create a new folder called station_data. Inside station_data, create folders rainfall, max_temp and min_temp:



Copy all rainfall, minimum temperature, and maximum temperature data from the extracted folder to their respective folders in station_data. This data is now ready to be inserted into the database.

Section 3: Creating environment files

As the repository is publicly available, .env must be created manually to ensure sensitive information such as the application's Mapbox API token are not exploited.

3.1 Root directory environment file

In the root directory, create a .env file with the following lines of code:

```
VITE_LOCAL_PORT=5173
VITE_DOCKER_PORT=5173
NODE_LOCAL_PORT=8080
NODE_DOCKER_PORT=8080
POSTGRES_LOCAL_PORT=5432
POSTGRES_DOCKER_PORT=5432
```

3.2 Client environment file

In client/, create a .env file with the following lines of code:

```
VITE_MAPBOX_ACCESS_TOKEN="YOUR ACCESS TOKEN"
VITE_SIGNUP='http://localhost:8080/api/signup'
VITE_AUTH_API_URL='http://localhost:8080/api/auth/'
VITE_USER_API_URL='http://localhost:8080/api/test/'
VITE_API_URL='http://localhost:8080/api'
```

3.3 Server environment file

In server/, create a .env file with the following lines of code:

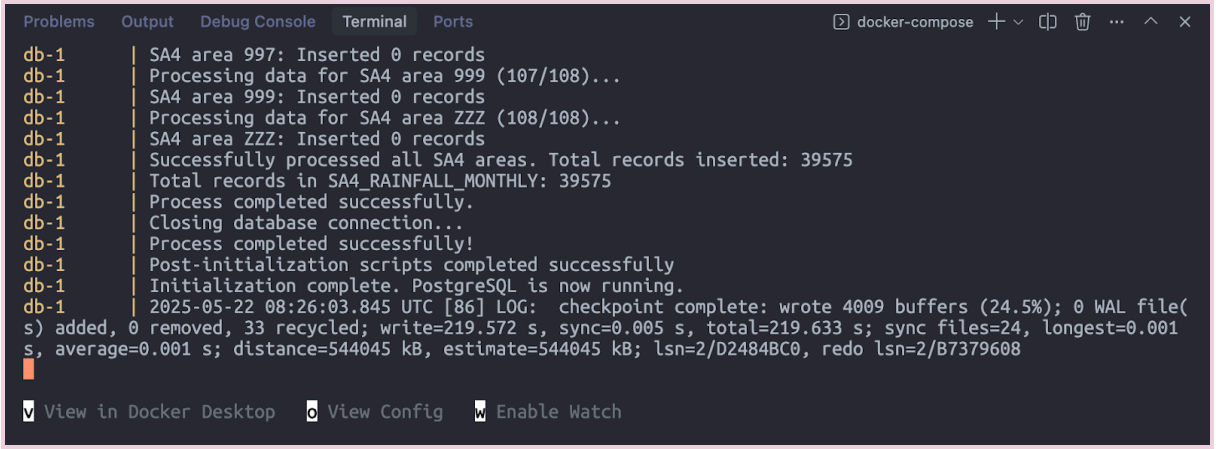
```
PGUSER=postgres
PGPASSWORD=postgres
PGHOST=db
PGPORT=5432
PGDATABASE=weather_db

# Nodemon settings
NODE_ENV=development
CHOKIDAR_USEPOLLING=true
```

Section 4: Building, starting and stopping the application

4.1 First start of the application

Using your terminal in the root directory, execute the following command:
`docker-compose up –build`
This will begin installing all necessary dependencies for the application, as well as inserting all relevant BOM data into the database. Please note this may take multiple hours to complete due to the sheer volume of data.
Once completed, you should see the following message in your terminal:



4.2 Restarting and stopping the application

To stop the application from running, while in the root directory, execute the following command in your terminal:

```
docker-compose down
```

To restart, execute:

```
docker-compose up
```

Do not use the `--build` flag to restart the application, as you will be required to re-initialise the entire database.

Section 5: Accessing the application

To access the application, proceed to the following link in your browser:

http://localhost:5173/

If the application has been built successfully, you will see the following page:

